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SPECIAL REPORT NO. 1

PROJECT GRUDGE

28 December 1951

AIR TECHNICAL INTELLIGENCE CENTER
WRIGHT-PATTERSON AIR FORCE BASE
DAYTON, OHIO

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This is a special report on the investigation of the sighting of an unidentified aerial object. Special reports such as this will be made on outstanding incidents and in incidents where such a report is requested by higher authority.

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FORT MONMOUTH, NEW JERSEY, INCIDENTS

On 10 and 11 September 1951, a series of both visual and radar sightings were reported from the Fort Monmouth, New Jersey, area.

I. VISUAL SIGHTING BY PILOT AND PASSENGER OF T-33 AIRCRAFT

A. Discussion

At approximately 1135 EDST an unidentified object was sighted by the pilot of a T-33 aircraft, an Air Force Lieutenant, enroute to Mitchell Air Force Base, New York, from Dover Air Force Base, Delaware. The object appeared to be over Sandy Hook, New Jersey, between 5000 ft. and 8000 ft. at 11 o'clock from the aircraft heading. The T-33 was approximately over Point Pleasant, New Jersey, at the time of the initial sighting. Upon seeing the object, the pilot started descending at 360° turn to the left in an attempt to intercept and identify the object. Approximately 45 seconds after the pilot first sighted the object, the passenger, an Air Force Major, who had been making a radio check, sighted the object. The object was then near Freehold, New Jersey, making a 120° turn toward the coast. The pilot continued his 360° turn but the object was lost as it crossed the coast. During the descending turn the speed of the T-33 increased from 450 to 550 mph and the altitude decreased from 20,000 ft. to 17,000 ft. (See inclosed overlay.)

When first sighted, the object appeared to be descending over Sandy Hook, New Jersey. It then leveled out and maintained a constant altitude. The object was round and silver in color but did not reflect the sunlight. At one time during the attempted intercept, it appeared flat. The size was judged to be 30 ft. to 50 ft. in diameter.

At approximately 1112 EDST, 10 September 1951, two balloons were released from the Evans Signal Laboratory, New Jersey, located at 40° 10' W and 74° 04' E. (See inclosed overlay.) These balloons are 7 ft. - 8 ft. in diameter at time of release and expand on ascending. They ascend at an average of 800 fpm and are painted silver for radar tracking. Experienced balloon observers state that when viewed from certain angles they appear to be disc-shaped. At 1135 EDST these balloons would have been at approximately 18,000 ft., and would have moved to a position nearly in line with Point Pleasant, New Jersey, and Sandy Hook. (Wind SSW at 10-15 knots.)

Attempts were made to use the information obtained from the interrogation of the T-33 crew and the data on the balloon launching to prove that the pilot and passenger of the T-33 had observed a balloon. However, not all of the data given was consistent with such a conclusion.

In an attempt to establish the fact that the object was a balloon, a flight path similar to the one given by the T-33 crew was assumed. (See "Assumed Path of T-33" in inclosure.) The T-33 crew was interrogated twice and gave different flight paths and tracts of the object at each one. It is therefore assumed that due to the altitude and speed of the T-33, and the fact

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that crew was intent on watching the object, they could not pin point their ground track any closer than 5 nautical miles and thus it would be feasible to assume a flight path within 5 nautical miles of the given track. Since the two interrogations as to location of the ground tracks differed to some extent, the track marked on a chart included with signed statement is assumed to be most nearly correct.

Referring to the assumed flight path on the inclosed overlay, at A, the object appeared to be over Sandy Hook. It will be noted that a comparatively small object closer to the a/c would appear to be large if assumed to be over Sandy Hook. (See Figure 1.)

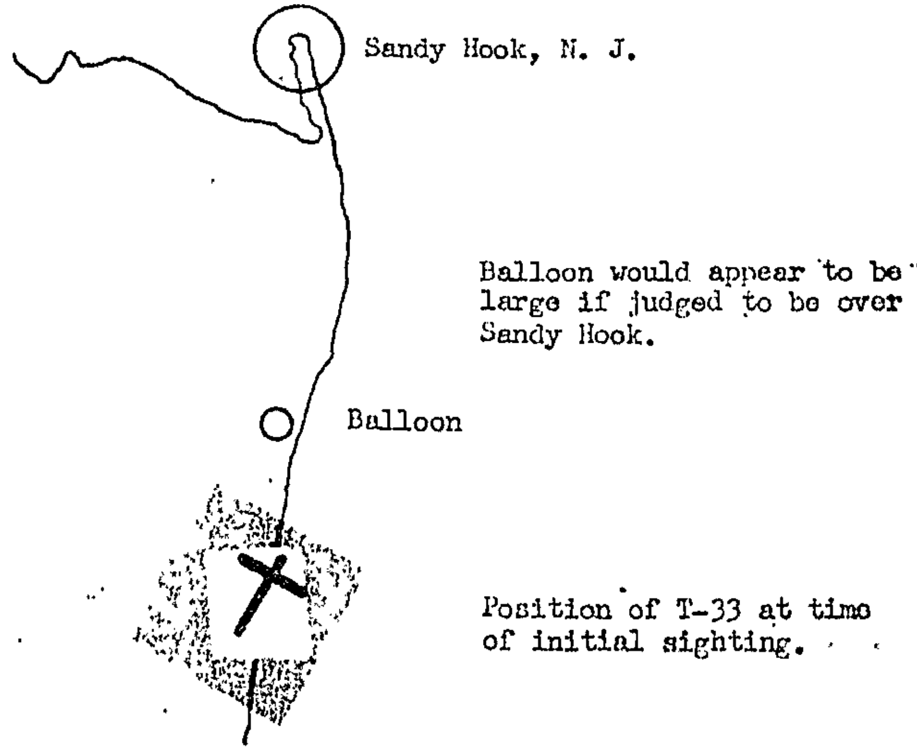


Figure 1. Plan View of Initial Sighting
(not to scale)

As the T-33 approached the balloon, the balloon appeared to be traveling at a high rate of speed. Several seconds must have passed after the initial sighting while the pilot decided that the object was not a conventional a/c and that he should attempt to identify it. During this period, it is assumed that the a/c continued on course making the object appear to be flying straight and level on a reciprocal heading. The fact that the object appeared to be descending when first sighted cannot be explained. The fact that only one of the two balloons was seen can be explained by the fact that the observers concentrated on one balloon and did not notice the other one.

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Forty-five seconds after the initial sighting, the passenger noted the object to be turning left near Freehold, New Jersey. This can be explained by the fact that the T-33 was turning and the relative motion caused the balloon to appear to be turning. As the T-33 continued inland, the line of sight changed until the balloon was silhouetted against the sea or sky and being silver blended into the background and was lost. This "disappearance" of balloons is a common occurence with pilots tracking research balloons.

It is apparent from the above that several assumptions had to be made in order to show that the object was one of the balloons released at Evans Signal Laboratory, but the fact there was a balloon in the near vicinity and the fact that the pilot and observer were not sure of their <u>exact</u> track adds a great deal of credence to the assumptions. However, since assumptions were made, it cannot be concluded that the object was definitely a balloon.

II. <u>RADAR SIGHTINGS FROM FORT MONMOUTH, NEW JERSEY</u>

A. Discussion

All of the radar sightings during this period were made by students at the Fort Monmouth training center. In addition to this, the students involved were taking a maintenance course. The instructor would put certain mechanical or electronic difficulty in the set and let the student find and remedy trouble. If the student became proficient in this phase, he was allowed to operate the set much the same as in tactical operations. No plotting records, logs or data of any type were kept. It should be stressed that these students were maintenance students, not operators.

1. On 10 September 1951 an AN/MPG-1 radar set picked up a fast-moving, low-flying target (exact altitude undetermined) at approximately 1110 hours southeast of Fort Monmouth at a range of about 12,000 yards. The target appeared to approximately follow the coast line changing its range only slightly but changing its azimuth rapidly. The radar set was switched to full-aided azimuth tracking which normally is fast enough to track jet aircraft, but in this case was too slow to be resorted to. The target was lost in the northeast at a range of about 14,000 yards.

Upon interrogation, it was found that the operator, who had more experience than the average student, was giving a demonstration for a group of visiting officers. He assumed that he was picking up a high-speed aircraft because of his inability to use full-aided azimuth tracking which will normally track an aircraft at speeds up to 700 mph. Since he could not track the target he assumed its speed to be about 700 mph. However, he also made the statement that he tracked the object off and on from 1115 to 1118, or three minutes. Using this time and the ground track, the speed is only about 400 mph.

No definite conclusions can be given due to the lack of accurate data but it is highly probable that due to the fact that the operator was giving a demonstration to a group of officers, and that he thought he picked up a very unusual radar return, he was in an excited state, accounting for his inability to use full-aided automatic tracking. He admitted he was "highly flustrated" in not being able to keep up with the target using the aided tracking. The weather on 10 September was not favorable for anomalous propagation.

2. On 10 September 1951, 1515 hours, an SCR 584, serial number 433, tracked a target which moved about slowly in azimuth north of Fort Monmouth at a range of about 32,000 yards at the extremely unusual elevation angle of 1350 mils, (altitude approximately 93,000 ft.). This was proven to be a weather balloon. It was tracked at the request of the Commanding Officer of the Student Attachment to determine the altitude in order to establish who won a pool concerning what the altitude of a balloon which was sighted might be.

3. On 11 September 1951, 1050 hours, two SCR 584's, serial number 217 and 315, picked up the same target northeast of Fort Monmouth at an elevation angle of 350 to 300 mils at a range of approximately 30,000 yards (approximate altitude 31,000 feet). The sets track automatically in azimuth and elevation and with aided range tracking are capable of tracking targets up to a speed of 700 mph. In this case, however, both sets found it impossible to track the target in range due to its speed and the operators had to resort to manual range tracking in order to hold the target. The target was tracked in this manner to the maximum tracking range of 32,000 yards. The operators judged the target to be moving at a speed several hundred miles per hour higher than the maximum aided tracking ability of the radar sets. This target provided an extremely strong return echo at times even though it was at maximum range, however, the echo signal occasionally fell off to a level below normal return. These changes coincided with manouvers of the target.

This sighting proved to be a weather balloon. How it was determined is unknown but ATIC was informed that it was a balloon by AFOIN-TC telecon TT-252, dated 5 October 1951, CSAF Item #12, which stated: "Radar sighting was later identified as weather balloon. Target track was vertical. Later exploded and descended to ground."

4. On 11 September 1951, at about 1330, a target was picked up on an SCR-584 radar set, serial number 315, that displayed unusual maneuverability. The target was approximately over Navesink, New Jersey, as indicated by its 10,000 yard range, 6,000 feet altitude and due north azimuth. The target remained practically stationary on the scope and appeared to be hovering. The operators looked out of the van in an attempt to see the target since it was at such a short range, however, overcast conditions prevented such observation. Returning to their operating positions the target was observed to be changing its elevation at an extremely rapid rate, the change in range was so small the operators believed the target must have risen nearly vertically. The target ceased its rise in elevation at an elevation angle of approximately 1,500 mils at which time it proceeded to move at an extremely rapid rate in range in a southerly direction once again the speed of the target exceeding the aided tracking ability of the SCR-584 so that manual tracking became necessary. The radar tracked the target to the maximum range of 32,000 yards at which time the target was at an elevation angle of 300 mils. The operators did not attempt to judge the speed in excess of the aided tracking rate of 700 mph.

It is highly probable that this is an example of anomalous propagation as the weather on 11 September was favorable for this type of phenomenon. The students stated that they were aware of this phenomenon, however, it is highly probable that due to the previous sightings of what they thought were unusual types of aircraft, they were in the correct psychological condition to see more such objects.

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III. CONCLUSIONS

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A. The unidentified aircraft reported by the T-33 pilots was probably a balloon launched by the Evans Signal Laboratory a few minutes before the T-33 arrived in the area.

B. The 1110 EDST radar sighting on 10 September 1951 was not necessarily a very high-speed aircraft. Its speed was judged only by the operator's inability to use aided tracking and this was possibly due to the operator being excited, and not the high speed of the aircraft.

C. The 1515 EDST radar sighting on 10 September 1951 was a weather balloon.

D. The 1050 EDST radar sighting on 11 September 1951 was a weather balloon.

E. The 1330 EDST radar sighting on 11 September 1951 remains unknown but it was very possible that it was due to anomalous propagation and/or the student radar operators' thoughts that there was a great deal of activity of unusual objects in the area.

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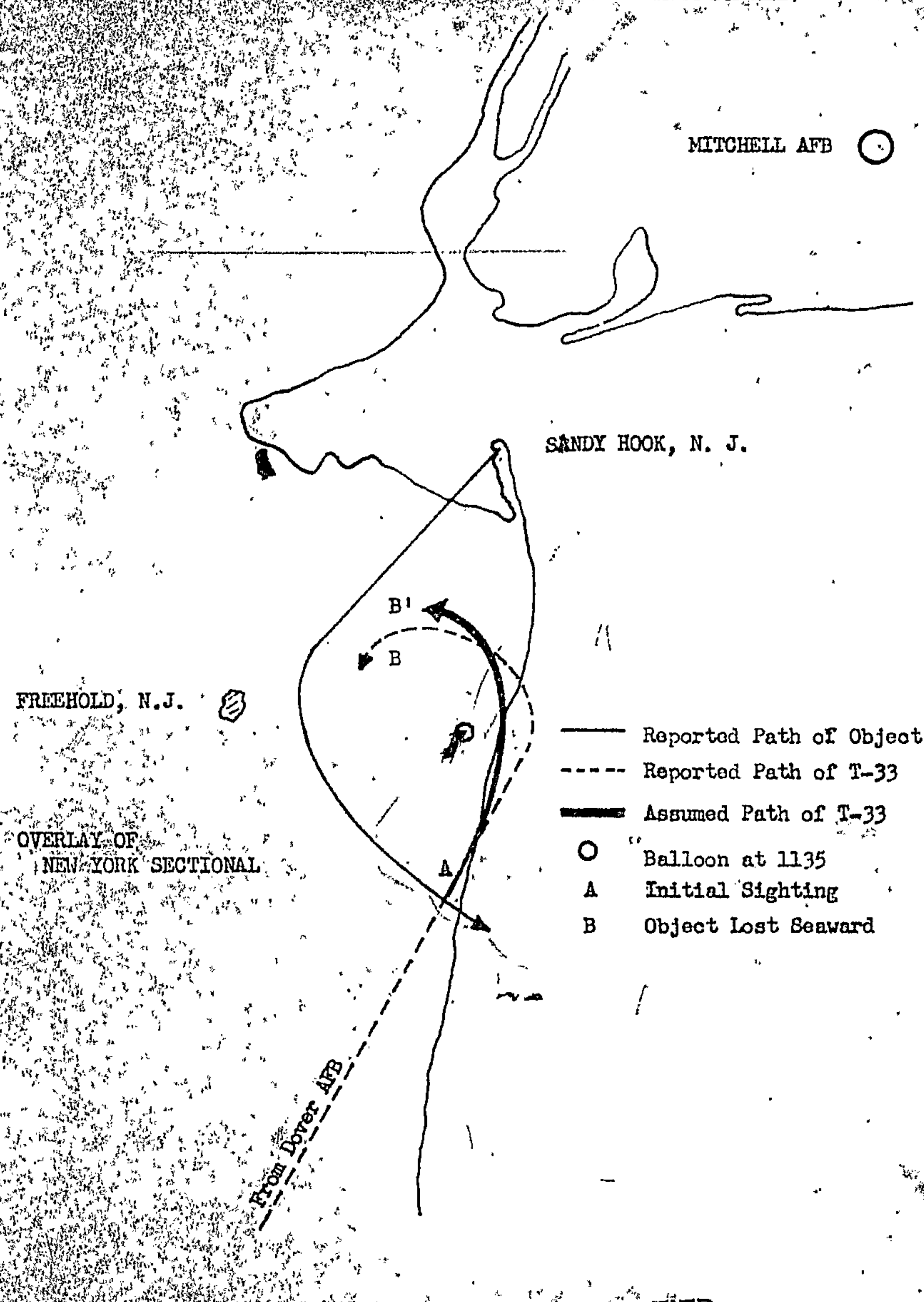


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